Planck-Wheeler length, time, and area, 426, 426n, 442, 446, 476n, 494, 494n, 518, 520, 563N55, 582N476
Plasma, 349f, 350, 353, 408–9, 408f
Plutonium, 199, 222–25, 222b, 573N220, 221
Polarization:
of light, 406n
of a gravitational wave, 393–94, 406n
of metal or a black-hole horizon, by electric charge, 405–7, 406f
Polchinski's paradox, 509–15, 510f, 512f, 513b–14b
Power notation for large numbers, 29b
Pressure:
physical cause of, 143–45
in rocks, 145, 201b
thermal, 144–45, 159–60
nonthermal, degeneracy 145–46, 148–52, 159–60, 169, 170b, 175–78, 193b, 200b–203b
nuclear, 177, 190–92, 194b-95b, 203f, 202b, 571N196
creates spacetime curvature, 118b–19b, 567N118–19
balances gravity in star, 135f, 137, 143–45, 144f, 151, 154f, 156b–57b, 160, 161f, 175–77, 190, 199, 202, 253
see also Equation of state
Probability, quantum mechanical, 147b, 477, 478f, 495f,496, 515
Pulsar:
nature of, 317–18
discovery of, 174, 468
as proof that neutron stars exist, 317
and tests of general relativity, 58, 393
radio waves from, 379
see also Neutron star

Quanta, 141
Quantum electrodynamics, 433
Quantum fields, 284
Quantum fields in curved spacetime, 436–39, 444b, 447–48, 482, 492, 517, 524
Quantum foam, 55–56, 477–78, 478f, 494–97, 495f
Quantum gravity:
Wheeler's motivation for, 210, 236–37, 244, 253–56, 262, 301, 449
Wheeler–DeWitt and Hartle–Hawking approach to, 479n
realm of validity, 85–86
nonexistence of time in, 476–77, 518–19
quantum foam of space (wormholes) in, 55–56, 477–78, 478f, 494–97, 495f
at endpoint of stellar implosion, 210, 476–79
in singularity inside black hole, 453, 476–79
and time-machine destruction, 518–20
what we may learn from in future, 52–25
Quantum limits on measurement accuracies, 372–75, 386
Quantum mechanics:
discovery of the laws of, 141, 147b, 162, 180–81
nature of and domain of, 50, 141, 372, 510–11, 514–15
as primary, with classical physics secondary, 514–15
and matter at high densities, 145–46, 147b, 150–52